

Your grade: 96.66%

Your latest: 96.66% • Your highest: 96.66% • To pass you need at least 80%. We keep your latest score.

Instructions

1.

You are working on a project using PyTorch and need to apply a 2D convolution to your input data. Which function should you use?

☐ torch.nn.Conv1d

☐ torch.nn.Linear

☒ torch.nn.Conv2d

☐ torch.nn.Conv3d

✔ Correct

Correct! The torch.nn.Conv2d function is used for applying 2D convolutions, which is essential for processing images and other 2D data.

1 / 1 point
2.

While designing a Convolutional Neural Network (CNN) for a new classification recognition task, you need to include various components. Which of the following components are crucial for building an effective CNN? (Select all that apply)

☒ Filters

✔ Correct

Correct! Filters are critical for detecting features in images, such as edges and textures.

☒ Pooling

✔ Correct

Correct! Pooling reduces the spatial dimensions, making computations more efficient and managing the representation size.

☒ Dropout layers

✘ This should not be selected

Incorrect. While dropout helps in regularization, it is not a key component specifically for CNNs.

☒ Fully connected layers

✔ Correct

Correct! Fully connected layers are essential for classifying the extracted features into final output classes.

☐ Recurrent layers

☒ Padding

✔ Correct

Correct! Padding is used to preserve the spatial dimensions of the image during convolution.

0.8 / 1 point

3.

In your CNN architecture, you are focusing on optimizing feature extraction. What is the primary function of filters in this context?

☐ To reduce the computational complexity by averaging pixel values.

☐ To normalize the pixel values and stabilize the training process.

☐ To convert the image from the spatial domain to the frequency domain.

☒ To detect specific features in the input images, such as edges or textures.

✔ Correct

Correct! Filters in CNNs are used to detect features like edges or textures, which are fundamental for learning hierarchical representations of the input data.

1 / 1 point

4.

You are optimizing a CNN for better performance on image classification tasks. What role does the pooling layer play in this optimization?

☐ To increase the resolution of the input image.

☐ To convert the image into a binary format for easier processing.

☐ To add non-linearity to the model.

☒ To reduce the spatial dimensions of the input, thereby decreasing the computational load.

✔ Correct

Correct! Pooling layers reduce spatial dimensions, making the network more efficient and less computationally demanding.

1 / 1 point

5.

During the training of a CNN, you notice that the training process is unstable. What is the main advantage of using batch normalization in this scenario?

☒ To stabilize the training process by normalizing the outputs of each layer

☐ To increase the learning rate during training

☐ To reduce the number of parameters and computational complexity

☐ To randomly drop units during training to prevent overfitting

✔ Correct

Correct! Batch normalization helps stabilize training by normalizing the outputs of each layer, leading to more consistent training dynamics.

1 / 1 point

https://www.coursera.org/learn/mastering-neural-networks-and-model-regularization/assignment-submission/tGzQh/exploring-convolutional-neural-networks-cnns/view-feedback

1/1